

8.6.2 Power control register 2 (PWR_CR2)

When the system is secure (TZEN =1), this register is protected against non-secure access when VDMSEC=1 in the PWR_SECCFGR register. A non-secure read/write access is RAZ/WI and generates an illegal access event.

When the system is not secure (TZEN=0), there is no access restriction.

This register is protected against non-privileged access when PRIV=1 in the PWR_PRIVCFGR register.

Address offset: 0x04

Reset value: 0x0000 0000

This register is reset when exiting the Standby mode.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	USV	IOSV	Res.	PVME4	PVME3	PVME2	PVME1	PLS[2:0]		PVDE	
					rw	rw		rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:11 Reserved, must be kept at reset value.

Bit 10 **USV**: V_{DDUSB} USB supply valid

This bit is used to validate the V_{DDUSB} supply for electrical and logical isolation purpose. Setting this bit is mandatory to use the USB peripheral. If V_{DDUSB} is not always present in the application, the PVM can be used to determine whether this supply is ready or not.

0: V_{DDUSB} is not present. Logical and electrical isolation is applied to ignore this supply.

1: V_{DDUSB} is valid.

Bit 9 **IOSV**: V_{DDIO2} Independent I/Os supply valid

This bit is used to validate the V_{DDIO2} supply for electrical and logical isolation purpose. Setting this bit is mandatory to use PG[15:2]. If V_{DDIO2} is not always present in the application, the PVM can be used to determine whether this supply is ready or not.

0: V_{DDIO2} is not present. Logical and electrical isolation is applied to ignore this supply.

1: V_{DDIO2} is valid.

Bit 8 Reserved, must be kept at reset value.

Bit 7 **PVME4**: Peripheral voltage monitoring 4 enable: V_{DDA} vs. 1.8 V

0: PVM4 (V_{DDA} monitoring vs. 1.8 V threshold) disable.

1: PVM4 (V_{DDA} monitoring vs. 1.8 V threshold) enable.

Bit 6 **PVME3**: Peripheral voltage monitoring 3 enable: V_{DDA} vs. 1.62V

0: PVM3 (V_{DDA} monitoring vs. 1.62V threshold) disable.

1: PVM3 (V_{DDA} monitoring vs. 1.62V threshold) enable.

Bit 5 **PVME2**: Peripheral voltage monitoring 2 enable: V_{DDIO2} vs. 0.9V

0: PVM2 (V_{DDIO2} monitoring vs. 0.9V threshold) disable.

1: PVM2 (V_{DDIO2} monitoring vs. 0.9V threshold) enable.